

Nirma University
Institute of Technology
Semester End Examination (IR), January-2025
Master of Computer Application, Semester - 1
6CS102CC22 Data Structures

Roll/
Exam No.
Time: 3 Hours

Supervisor's Initial
with date

Max Marks: 100

Instructions:

1. Attempt all questions. (No Optional Questions)
2. Use section-wise separate answer book.
3. Figures to the right indicate full marks.
4. Draw neat sketches wherever necessary.
5. Assume suitable data wherever necessary and mention the same.
6. Use of a scientific calculator is permitted.

SECTION - I

Q-1. Answer the following questions. [16]

(A) Differentiate between Linear Data Structure and Non-linear Data Structure. (4)
CLO1
BL4

(B) For array A of dimension 4X5 with base address 2500 and element memory size of 4 bytes, find the address of element A[3][4] using row-major order and column-major order. Consider the lower bound for row and column is 1. (6)
CLO1
BL3

(C) Represent the polynomial given below using a singly linked list and explain the node format for the same. Write an algorithm to display a polynomial represented using a singly linked list. (6)
CLO1
BL3

Polynomial: $25X^3Y^2 + 30X^2Y + 14XY^4 + 4X + 8Y + 3$

Q-2. Answer the following questions. [18]

(A) Convert the following infix expression into a postfix expression. (6)
CLO2
BL3
Solve the resultant postfix expression using stack.
 $5 + 3 - (4 - 2) * 9 / 6$

(B) A circular queue is more efficient than a standard queue (simple queue) when the queue data structure is implemented using static memory allocation. Justify this statement using examples. Also, write an algorithm to insert an element in a standard queue (simple queue) implemented using static memory allocation. (6)
CLO2
BL5

(C) Formulate the algorithms to perform the following operations: (6)
CLO2
BL6
(i) Insert a node at the front in the Singly-linked list
(ii) Delete a node from last in the Singly-linked list

Q-3. Answer the following questions. [16]

(A) Formulate an algorithm for the binary search method for searching an element from the given list of sorted elements. (4)
CLO3

BL3

(B) Show step-by-step tracing to sort the following data in ascending order using bubble sort. (6)
CLO3

BL3

52 36 93 61 87 44

(C) Write an algorithm to perform Quick Sort. Discuss the time complexity for the same. (6)
CLO3

BL3

SECTION - II

Q-4. Answer the following questions. [16]

(A) Write Breadth-First Search traversal and Depth-First Search traversal for the graph whose adjacency list is given below. Start traversal from vertex A. (8)
CLO2

BL3

Vertex	Adjacent Vertices
A	B C D
B	A E F
C	A G H
D	A G I
E	B J
F	B J K
G	C D K L
H	C L
I	D L
J	E F
K	F G
L	G H I

(B) Construct Binary Tree from given pre-order traversal and in-order traversal. Show step-by-step construction of the binary tree. (8)
CLO2

BL3

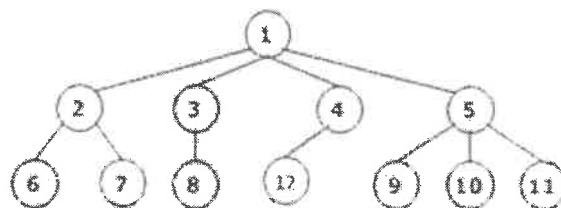
Pre-order: L E B H N R P

In-order: B E H L N P R

Q-5. Answer the following questions. [18]

(A) Convert the following general tree into an equivalent binary tree. (6)
CLO4

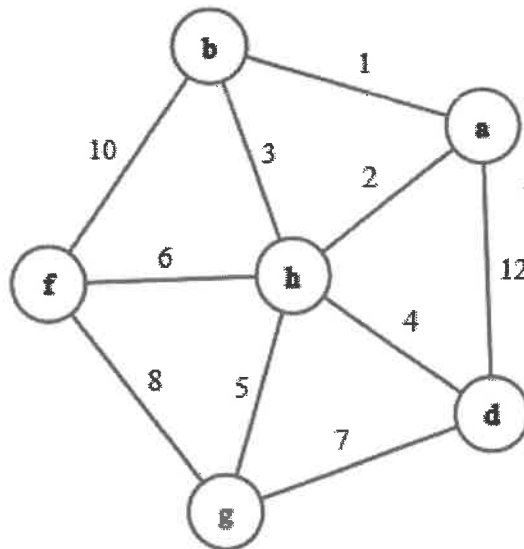
BL3



- (B) For graph G defined below, find the in-degree, out-degree and total degree of each vertex. Also, write the adjacency matrix and adjacency list for this graph. Determine whether graph G is a directed graph or an undirected graph. (6)
- CLO4
BL3

$G(V, E)$ where $V = \{a, b, c, d, e\}$ and $E = \{(a, b), (a, e), (b, d), (b, e), (c, a), (c, d), (c, e), (d, c), (e, d)\}$

- (C) Find the minimum cost spanning tree (MST) using Kruskal's algorithm for the graph given below and also calculate the minimum cost for the same. (6)
- CLO4
BL3



- Q-6. Answer the following questions. [16]

- (A) Create a Binary Search Tree (BST) for the following data. Write in-order, pre-order and post-order traversal of generated BST. (8)
- CLO3
BL3

40 82 25 33 62 18 74 59

- (B) Critically distinguish between B-Tree and Binary Search Tree. (8)
- CLO3
BL4
- Construct a B-tree of order 5 by inserting the following numbers in the given sequence.

11 35 77 48 121 344 230 165 150 55 81